

Candidate Name

Centre Number

Candidate Number



ZIMBABWE SCHOOL EXAMINATIONS COUNCIL
General Certificate of Education Ordinary Level

PHYSICS

PAPER 2 Theory

4023/2

2 hours 15 minutes

NOVEMBER 2024 SESSION

Additional materials:
Electronic calculator
Answer booklet

INSTRUCTIONS TO CANDIDATES

Write your name, centre number and candidate number in the spaces at the top.

At the end of the examination fasten the answer booklet used securely to the question paper.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets [] at the end of each question.

This question paper consists of two compulsory sections, **A** and **B**.

FOR EXAMINER'S USE

Section A	
Section B	9
	10
	11
	12
Total	

This question paper consists of 14 printed pages.
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Section A

Answer all questions.

- 1 (a) (i) Define *momentum*.

_____ [1]

- (ii) State the SI units of momentum.

_____ [1]

- (iii) State how the acceleration is related to the force applied.

_____ [1]

- (b) A car of mass 800 kg travelling at 20 m/s, decelerates to a speed of 11 m/s.

Calculate the change in momentum.

[2]

2 (a) A boy releases an arrow at a force of 120 N and hits a tree 8 m away.

(i) Calculate the work done by the arrow.

[2]

(ii) Calculate the power produced if the arrow takes 4 s to reach the tree.

[2]

(iii) State the energy in the stretched bow.

[1]

- 3 (a) (i) State **one** material used for constructing large structures.

_____ [1]

- (ii) Give **one** advantage of using the material suggested in **a(i)**.

_____ [1]

- (b) Explain why

- (i) large dams are usually arch shaped,

_____ [1]

- (ii) dam walls are made of reinforced concrete.

_____ [2]

- 4 The stone in **Fig 4.1** has a mass of 5.0 kg. The work done in moving the stone up the smooth slope from X to Y is 150 J.

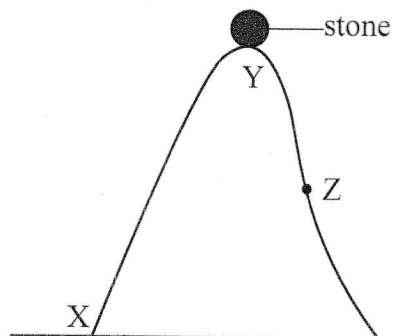


Fig 4.1

- (a) (i) Define *work done*.

[1]

- (ii) Determine the potential energy of the stone at Y.

[1]

- (b) Calculate the speed of the stone at Z, the midpoint of its fall.

[3]

- 5 (a) Define

- (i) *boiling*.

[1]

- (ii) *melting*.

[1]

- (b) Calculate the amount of heat energy needed to change 10 g of water from 1.0°C to 70.0°C .
{specific heat capacity of water is $4.2\text{ J/g}^{\circ}\text{C}$ }

[3]

- 6 (a) State any **one** use of electricity in the home.

[1]

- (b) Give any **one** safety precaution when using electrical gadgets in the home.

[1]

- (c) A boiling kettle labelled 220 V, 5 A was used for 5.0 minutes to boil water.
Calculate the electrical energy it used to boil the water.

[2]

- (d) Explain why fuses are always on live wire.

[1]

- 7 (a) (i) Fig 7.1 shows a transformer with 1200 turns in the primary coil and 2400 turns in the secondary.

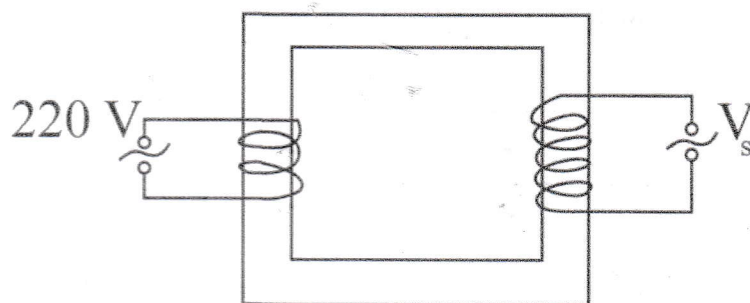


Fig 7.1

Name the type of transformer shown in Fig 7.1.

[1]

- (ii) Calculate V_s .

[2]

(iii) Describe and explain **one** way of improving the efficiency of a transformer.

[2]

8 (a) Explain what is meant by *thermionic emission*.

[1]

(b) **Fig 8.1** shows an electron entering a uniform magnetic field which is directed into the paper.

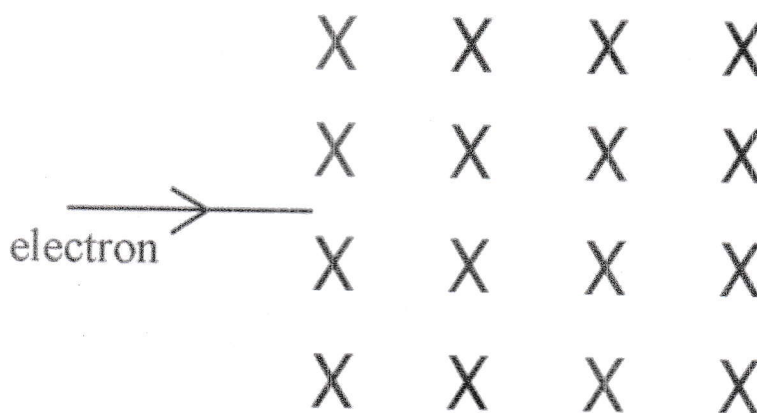


Fig 8.1

On **Fig 8.1**, complete the path followed by the electron as it enters and leaves the uniform magnetic field.

[3]

(c) Give **one** property of electrons.

[1]

Section B

Answer **any three** questions from this section.

- 9 (a) Give a molecular account of how heat is transferred when a metal is heated at one end. [3]

- (b) Fig 9.1. shows a car of mass 900 kg travelling at a steady speed of 20 m/s against a resistive force of 1000 N.

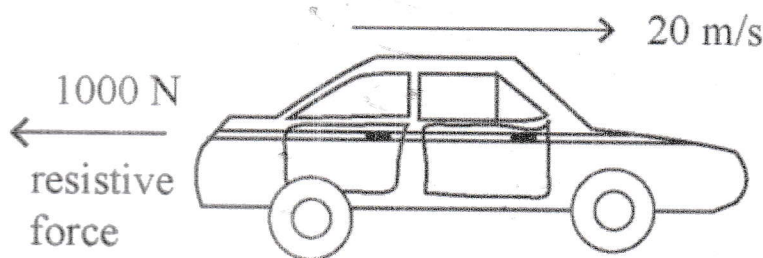


Fig 9.1.

- (i) Name a form of energy in the fuel, used by the engine. [1]
- (ii) Calculate the
 1. kinetic energy of the car,
 2. energy used in 2 s against the resistive force. [4]
- (iii) Determine the minimum power that the car engine has to deliver to the wheels for the car to just move forward. [2]
- (iv) Suggest which other energy conversion occur besides chemical to kinetic energy. [2]
- (c) (i) State the method used to join the
 1. steel door lock and wooden door,
 2. electric socket and concrete wall. [2]
- (ii) Describe, with reasons, how dam walls are constructed. [3]

- (iii) Fig 9.2. shows a roof truss.

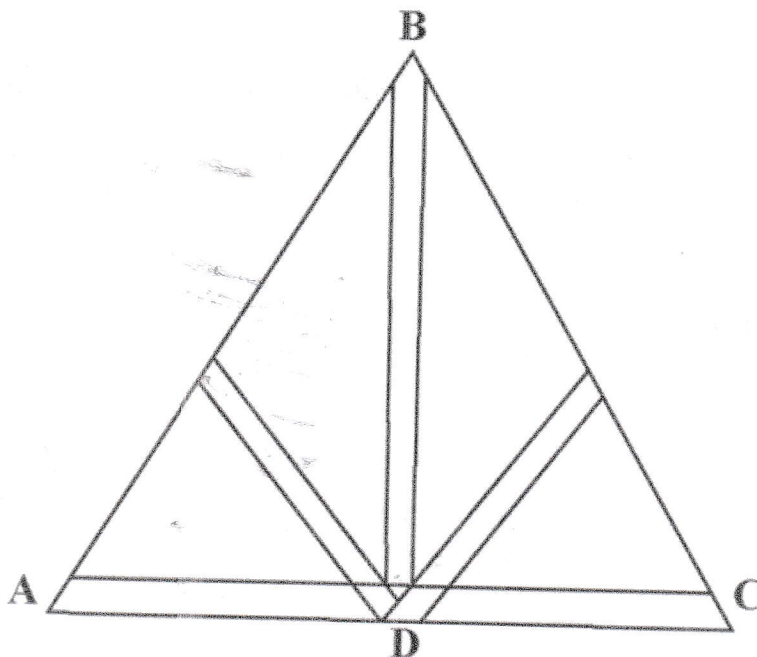


Fig 9.2

Identify which beam(s) are ties and struts.

[3]

- 10 (a) (i) State any **two** practical machines that use the principle of moments. [2]

- (ii) Fig 10.1 shows a flat plate which weighs 100 N and used to lift loads. The plate is lifted upwards using a handle with a force of 30 N to keep the handle horizontal. The point marked M is the centre of mass of the plate and A is the pivot.

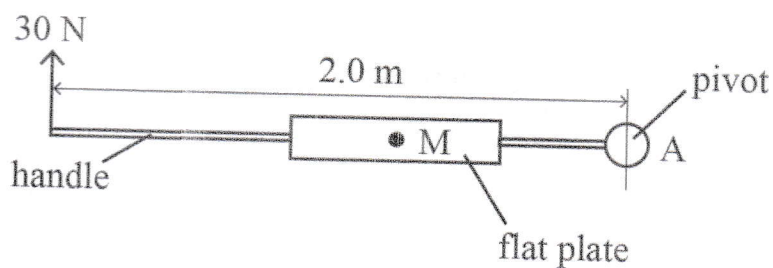


Fig 10.1

1. Copy Fig 10.1 and draw an arrow that shows one of the forces acting on the plate.
2. Determine the moment of the 30 N force about the pivot.
3. Calculate the distance between A and M.

[4]

- (iii) If the plate is loaded with 200 N at M, find the effort needed to lift the load. [2]

- (b) Fig 10.2 shows a rectangular tank of water.

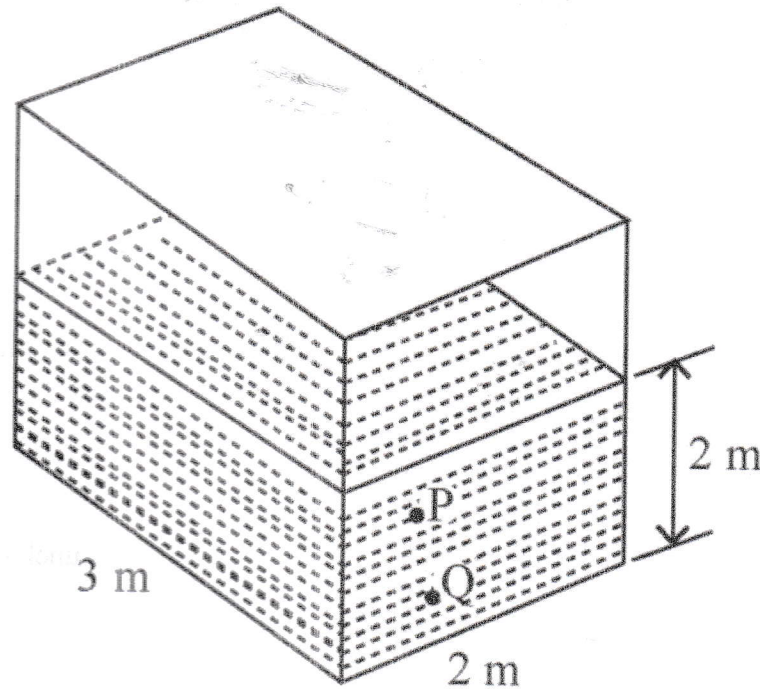


Fig 10.2

- (i) Calculate the weight of the water in the tank if the density of water is 1000 kg/m^3 and $g = 10 \text{ m/s}^2$. [2]
- (ii) Calculate the pressure of the liquid at the bottom of the tank. [2]
- (iii) State and explain which object P or Q experiences higher pressure. [2]
- (c) A rectangular bill board has an area of 4.0 m^2 . The force exerted by air pressure on one side of the board is 60 000 N.
- (i) Calculate the pressure exerted on the bill board by the wind. [2]
- (ii) If a 1.0 m^2 bill board is put near the billboard in c(i), state and explain the likely impact of this amount of wind force. [2]

- (iii) Use a diagram to describe the construction of a simple manometer. [2]

- 11 (a) (i) Define *refractive index*. [1]

- (ii) A ray of light in air is incident on an air-water boundary at angle of 40° .
Calculate the angle of refraction of the ray, if the refractive index of water is 1.33. [2]

- (iii) Describe, with the aid of well labelled diagram(s), *total internal refraction*. [4]

- (b) Fig 11.1 shows an isolated metal sphere labelled A, placed on an insulated holder.

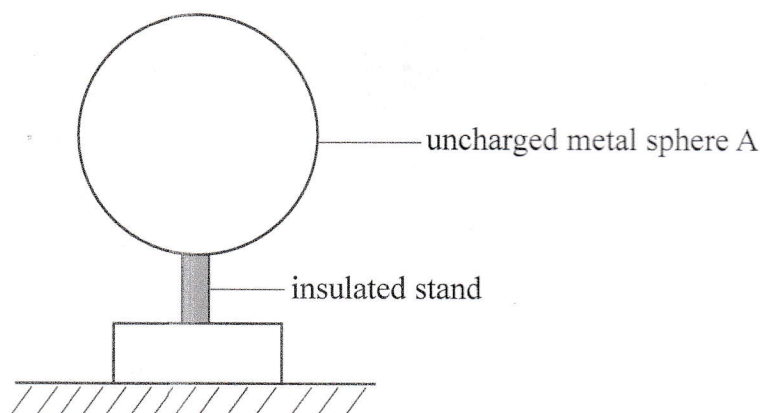


Fig 11.1

A negatively charged metal sphere is brought closer to sphere A but not in contact.

Describe and explain, with the aid of a diagram,

- (i) the distribution of charge in sphere A. [2]
- (ii) what happens when sphere A is connected to the earth. [2]
- (iii) how charge is distributed in sphere A when the earth in (ii) is removed first and then the negatively charged sphere later. [2]

- (c) (i) Explain what is meant by a *magnetic field*. [1]
- (ii) Describe how a magnetic material can be magnetised by the method of stroking. [4]
- (iii) State any **two** applications of permanent magnets. [2]
- 12 (a) (i) State three factors that affect the magnitude of induced electromotive force. [3]
- (ii) **Fig 12.1** shows a solenoid carrying current.

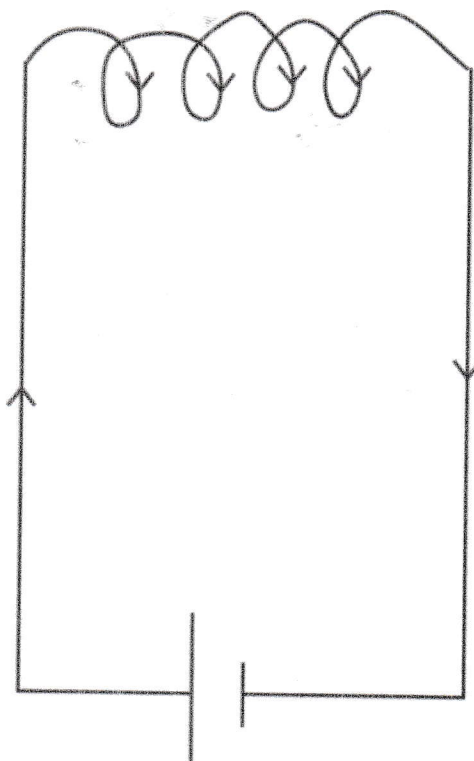


Fig12.1

1. Name the rule that is used to predict the direction of the magnetic field due to a solenoid.
2. Copy **Fig12.1** and draw the magnetic field pattern around the solenoid.
3. Describe and explain what happens if alternating current is used instead of direct current. [5]

- (b) (i) Draw circuit symbols of the following electronic components:
1. thermistor
 2. light dependent resistor
 3. capacitor
 4. magnetic relay switch
- [4]
- (ii) Give **two** uses of a capacitor.
- [2]
- (c) An isotope of carbon is represented by ${}^{14}_6\text{C}$.
- (i) Define the term *isotope*.
- [2]
- (ii) Calculate the number of neutrons in ${}^{14}_6\text{C}$.
- [2]
- (iii) Describe one use of Carbon 14 isotope.
- [2]